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The Use of Atomic Fluorescence Spectrometry for Mercury Determination in the Petrochemical Industry

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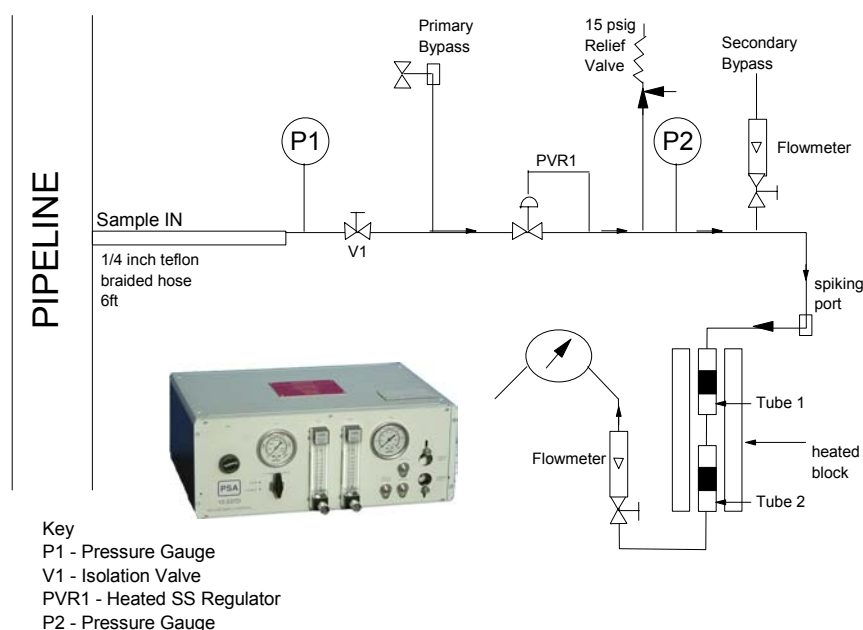
Introduction

Knowledge of the mercury content in petrochemicals is extremely important. Firstly, mercury is highly toxic and is of environmental concern and secondly, the damage caused to petrochemical plants can be financially crippling especially when unscheduled shutdowns are forced. Mercury has been found to be responsible for many cases of selective hydrogenation catalyst deactivation. Palladium based catalysts are commonly used for the selective hydrogenation of acetylenic species in the steam cracking of C_2 to C_4 cuts. Mercury is known to be the cause of corrosion problems with aluminum-based heat exchangers, rotors and condensers at natural gas refinery plants. Heat exchanger replacement is a costly operation due to the capital investment of the exchanger itself and the plant down time incurred for its replacement. This paper describes how Atomic Fluorescence Spectrometry (AFS) can be applied to the measurement of mercury in petrochemical samples such as natural gas, LPG, LNG, naphtha, condensates and crude oil. Performance characteristics of the analysers will be discussed with reference to sampling, calibration, accuracy, precision and long term reliability.

Mercury in Natural Gas and LPG

The offline sampling of high pressure natural gas and liquefied gas streams such as LPG is achieved using the portable PSA 10.537D (Figure 1).

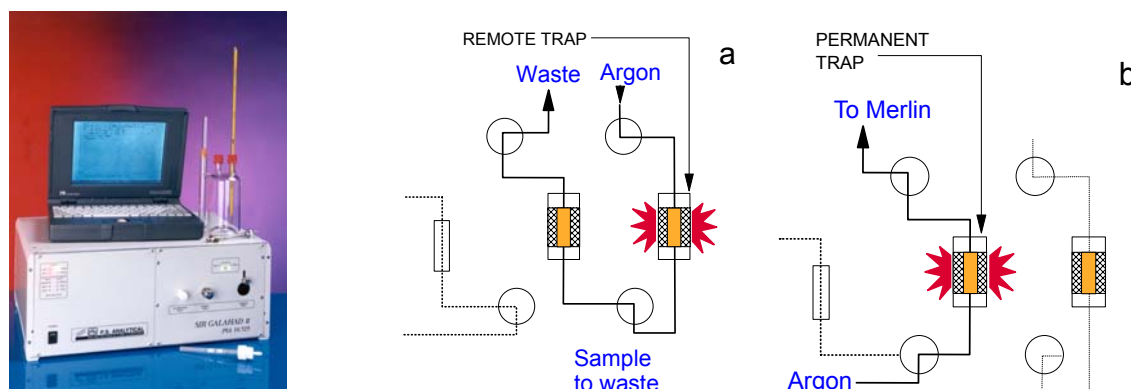
Figure 1 Schematic of PSA 10.537D Offline Sampling System in Series Mode



The unit has been designed to operate in hazardous Zone 1 areas using ExDIICT3 rated components. This enables the samples to be collected close to the pipeline and therefore minimizes any contamination or losses often found when sampling for mercury. In normal operation the sampling tubes are used in series to test for breakthrough of mercury. The system can be modified in the field so that the sample can be collected in parallel. To overcome condensation of hydrocarbons on the sampling tubes, the adsorbent AmasilTM was maintained at 140°C, which is well above the dew-point of the hydrocarbon gas.

Different sample volumes were collected to ensure that no matrix interference is encountered. Samples may also be collected in parallel mode. The sample tubes were analysed by dual amalgamation – atomic fluorescence spectrometry (PSA 10.525). The measurement time is approximately 5 minutes. This unit is shown in Figure 2.

Figure 2 Photograph and Schematic of PSA 10.525 Sir Galahad



A typical application for Hg in Natural Gas would be to monitor the inlet and outlet of Hg removal beds. Table 1 presents typical data from a gas plant in the USA. Mercury removal based on carbon impregnated with sulphur reduces the Hg to below 16ng/m³. Excellent correlation between the sample volume collected and the concentration measured was found indicating no matrix interference. Method detection limit for Hg in natural gas is 0.1ng/m³. AFS has a linear dynamic range of 7 orders of magnitude. In this application the volume of sample collected can be reduced so that an upper working range of 3000000ng/m³ is achieved. Table 2 presents some typical data using PURASPEC™ on a gas plant in the UK. The mercury removal bed was found to decrease the result from around 67000ng/m³ to 15ng/m³.

Table 1 Typical results before and after Hg removal beds using activated carbon impregnated with sulphur

Sample Reference	Tube 1 (ng)	Tube 2 (ng)	Volume (l)	Conc (ng/m ³)
Bed Inlet	7.547	0.072	1	7619
Bed Inlet	7.693	0.076	1	7769
Bed Inlet	15.042	0.121	2	7582
Bed Inlet	38.213	0.218	5	7686
Bed Outlet	0.214	0.025	20	12
Bed Outlet	0.742	0.068	50	16
Bed Outlet	1.358	0.105	100	15
Bed Outlet	6.985	0.098	500	14

Table 2 Typical results before and after Hg removal beds using PURASPEC

Sample Reference	Tube 1 (ng)	Tube 2 (ng)	Volume (l)	Conc (ng/m ³)
Bed Inlet	63.23	0.03	1	63260
Bed Inlet	65.59	0.01	1	66000
Bed Inlet	71.89	0.01	1	72000
Bed Inlet	132.45	0.03	2	66240
Bed Outlet	1.308	0.077	90	15.4
Bed Outlet	1.101	0.043	75	15.4
Bed Outlet	3.957	0.779	313	15.1
Bed Outlet	0.701	0.093	50	15.9

The measurement can also be performed online for multiple streams using the same approach. The online analyser enclosure and sampling system is shown in Figure 3 along with a schematic diagram of the system. To comply with safety regulations the cabinet was continuously purged with air. The power to the analyser is only applied after the purge conditions are maintained and uninterrupted. The sampling system has numerous safety features such as pressure switches and solenoid valves. In a fault condition such as regulator failure the sample is automatically isolated by the pressure switch activating the solenoid. A schematic diagram of the PSA sampling is shown in Figures 4. Typical data monitoring the outlet of a Hg removal bed over a 3 month period is shown in Figure 5.

Figure 3 Photograph of Online System for Hg in Natural Gas

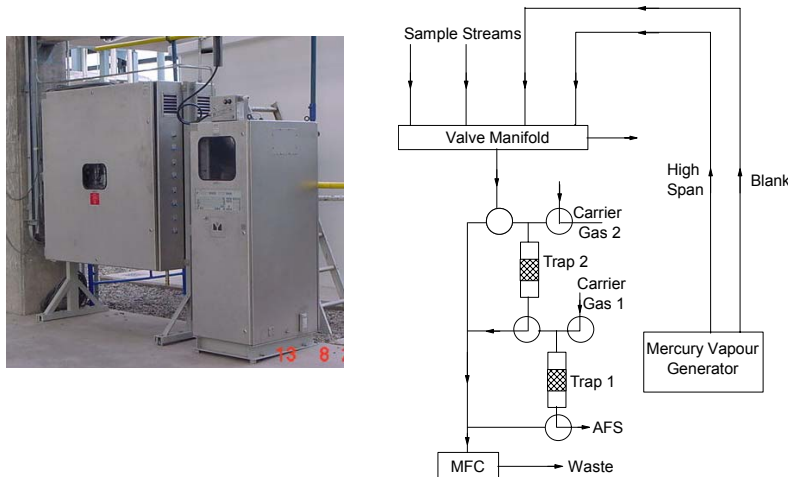


Figure 4 Schematic Diagram of Online Sampling System for a single stream

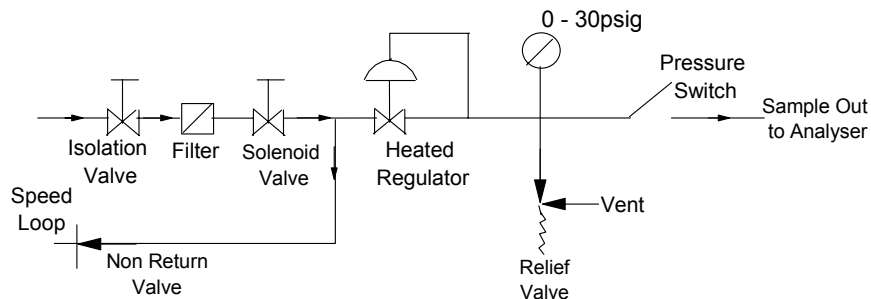
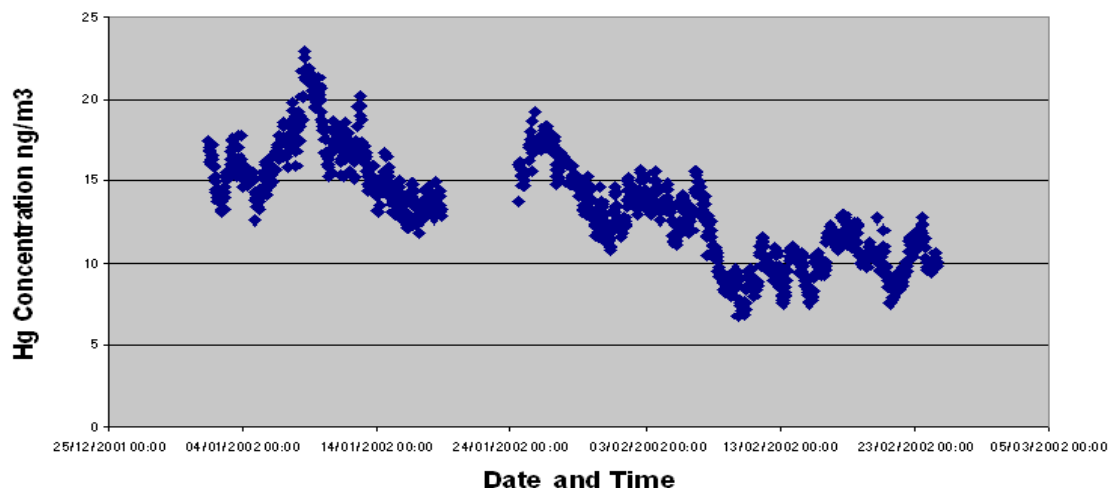


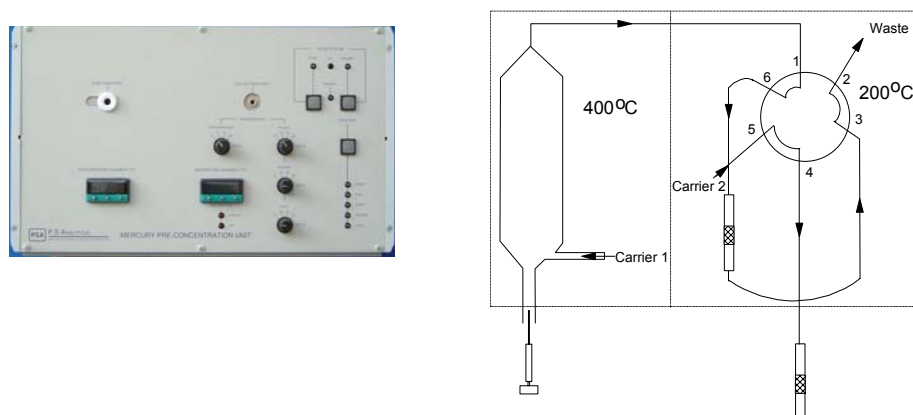
Figure 5 Hg Concentration Trend for Outlet of Bed



Hg in Naphtha and Natural Gas Condensates

The PSA 10.515 Hg Preconcentration is specifically designed to determine Hg in naphtha and natural gas condensates. Samples are injected directly without sample preparation into a high temperature vaporization chamber. The volatilized sample is then transferred using an inert carrier gas to a heated Amasil™ trap. Mercury compounds are collected whilst the vaporized hydrocarbon flows to waste. The trapped Hg is then released by thermal de-sorption and subsequently analysed using the 10.525 Sir Galahad (figure 6). The system offers numerous advantages compared to wet digestion or pyrolysis methods. These include a 10 minute measurement time and large injection volumes providing method detection limits around 0.02ppb by weight. The combination of analyte pre-concentration using Amasil™ and AFS provides excellent selectivity. Table 3 shows typical data for a Malaysian condensate containing 23.2 ± 1.3 ppb by weight spiked with different Hg species. Although the mercury speciation in this sample was unknown the spike and recovery experiments indicate that a good total mercury measurement was obtained.

Figure 6 Schematic and photograph of PSA 10.515 Preconcentration system for Hg in Condensates and Naphtha



One of the main advantages of the volatilization – AFS technique is the superior method detection limits achievable. Table 4 presents a method detection limit comparison. The method detection limit (MDL) for wet digestion is dependent on reagent purity, dilution factors and the sensitivity of the detection method being employed. The MDL for Pyrolysis AAS is poor. This is because of the limited amount of sample that can be pyrolysed in a safe manner, the high and variable blanks from sample boats and additives and the difficulty in transferring and weighing a volatile sample.

Naphtha below 1ppb (weight) can be sold at a premium. To achieve detection limits below this figure the Volatilization AFS is the only approach that can ensure a reliable measurement at this level. A Mercury removal process from WCP International using PURASPEC adsorbent was found to produce naphtha with Hg content well below 1ppb.

Table 3 Typical data for a Malaysian Condensate

No.	Mercury species added	% Recovery $\pm$ S.D.		
		Concentration		
		10 ppb	30 ppb	50 ppb
1	Dimethylmercury	111 $\pm$ 13	109 $\pm$ 14	105 $\pm$ 9
2	Mercury Chloride	100 $\pm$ 11	98 $\pm$ 13	98 $\pm$ 3
3	Ethylmercury chloride	97 $\pm$ 10	99 $\pm$ 8	96 $\pm$ 4
4	Methylmercury chloride	102 $\pm$ 4	92 $\pm$ 5	92 $\pm$ 8
5	Diethylmercury	113 $\pm$ 8	92 $\pm$ 5	99 $\pm$ 5
6	Equal mixture of above	90 $\pm$ 5	88 $\pm$ 2	97 $\pm$ 1
7	Conostan CRM	101 $\pm$ 3	96 $\pm$ 4	98 $\pm$ 2

Table 4 Comparison of Method detection limits

Technique	Max Sample (Grams)	Min Dilution (mL)	MDL ppb (weight)
CV-AAS	1.0	50	5.0
CV-ICPMS	1.0	50	2.0
CV-AFS	1.0	50	0.2
Pyrolysis AAS	0.1	None	2.0
Volatilisation AFS	1.0	None	0.02

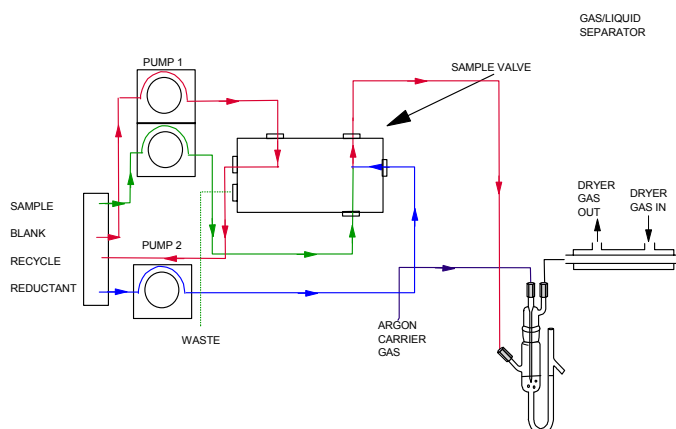
Mercury Fractionation in Crude Oil

An automated continuous flow vapour generation system (PSA 10.025) coupled to AFS was used to determine Hg in crude oil (Figures 7 & 8).

Figure 7 10.025 Millennium Merlin System



Figure 8 Schematic representation of the PSA 10.025 Millennium Merlin System



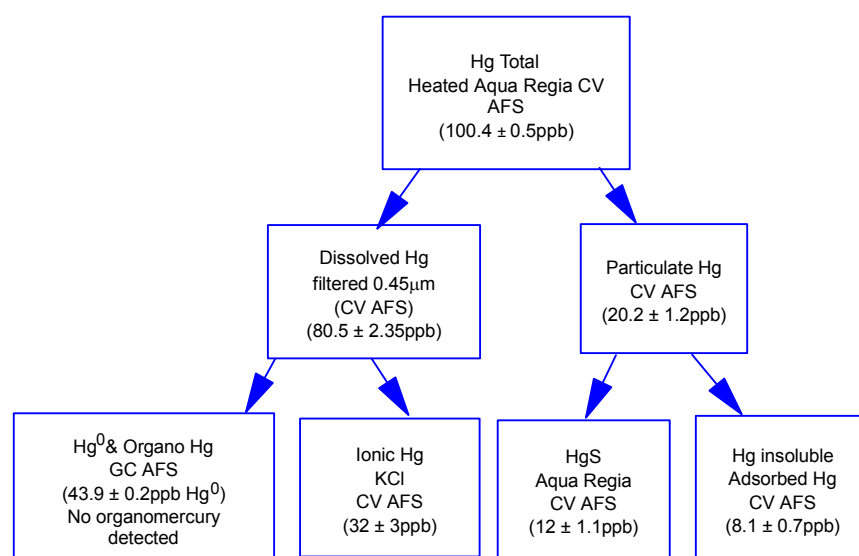
For the determination of total Hg, the sample was initially homogenized by shaking for 1 hour and ultrasonicated for 15 minutes. The sample was then heated in a reflux arrangement with aqua regia at 120°C for a period of one hour. Typical data for total Hg in crude oil is shown in Table 5.

All samples were prepared in duplicate and analysed in triplicate. With each batch of samples a NIST certified Conostan Reference standard was prepared in base oil at 34.5ng/ml. Samples were also spiked with this material to check for matrix interferences.

Table 5 **Total Mercury in Crude Oil and Condensate Samples**

Sample	Prep	Mercury (ng/ml)	Mean Result (ng/ml)
Conostan CRM	1	35.3 ± 0.3	34.5 ± 0.4 (100%)*
	2	33.7 ± 0.3	
North Sea Oil 1	1	97.1 ± 0.2	100.4 ± 0.5 (97.2%)*
	2	103.6 ± 0.5	
North Oil Sea 2	1	78.5 ± 0.1	76.4 ± 2.9 (102.1%)*
	2	74.3 ± 0.3	
Thai Crude Oil	1	2896 ± 15	2944 ± 7 (96.4)*
	2	2991 ± 8	
Venezuelan Crude Oil	1	3852 ± 18	3889 ± 2 (95.1)*
	2	3926 ± 10	
Algerian Condensate	1	733 ± 14	737 ± 21 (98.2)*
	2	741 ± 15	
* Spike Recovery Data			

Particulate mercury was established by filtering through a 0.45 micron filter and digesting/extracting the filter medium. In the particulate mercury fraction we were able to distinguish between mercury sulphide and insoluble mercury. This was done using high temperature extraction with aqua regia and nitric acid respectively. In the latter case mercury sulphide was not digested and therefore the mercury sulphide content was obtained by difference. Ionic mercury was extracted using a saturated solution of potassium chloride. Organomercury and elemental Hg was determined using a specially designed capillary GC- atomic fluorescence spectrometer after a direct injection of the filtered samples. Figure 9 summarises the fractionation in terms of flowchart for North Sea Crude Oil.

Figure 9 **Procedural Flowchart for Hg Fractionation in Crude Oil**

Conclusions

AFS is a sensitive and versatile detection system for mercury in petrochemical applications. Numerous systems have been installed around the world and have been operating for several years. More recently, a method to establish the fractionation of Hg in crude oil has been developed and excellent mass balance was obtained.